

Lab 2 — Detecting Network Attacks with Suricata

➤ Learning Objectives

By the end of this lab, students should be able to:

- Analyze network traffic using **PCAP files**
 - Detect attacks using **Suricata rules**
 - Generate and observe **real-time attacks**
 - Correlate **alerts with network activity**
 - Develop a **security analyst mindset**
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➤ Prerequisites

Students must have:

- Basic understanding of Suricata configuration and rules (Lab 1)
- Access to:
 - A Linux machine (or VM)
 - Test PCAP files
- Basic networking knowledge (IP, ICMP, TCP)

Optional but recommended:

- Wireshark
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➤ Lab Scenario

You are part of a **Security Operations Center (SOC)** team.

Your task is to:

- Analyze suspicious traffic
- Detect malicious activity
- Identify attackers using Suricata

You will complete two types of detection:

1. **Offline analysis (PCAP)**
2. **Live attack detection**

Part 1: Offline Analysis (PCAP Investigation)

Mission 1: Analyze a PCAP File

You are given a network capture file containing suspicious activity:

<https://www.malware-traffic-analysis.net/>

Download any PCAP file from recent exercises (prefer small size < 50MB)

Tasks

- Load the PCAP file into Suricata
- Generate alerts based on the ruleset
- Identify at least **2 suspicious events**

Deliverables

- Number of alerts generated
- Type of detected activity
- Source and destination IPs

Mission 2: Identify the Attack

Analyze the alerts generated.

Tasks

- Determine the **type of attack** (scan, flood, brute force, etc.)
- Identify:
 - Attacker IP
 - Target IP
 - Protocol used
- Explain the behavior observed

Deliverables

- Short analysis (5–10 lines)
- Evidence from alerts/logs

Part 2: Live Attack Detection

Mission 3: Run Suricata in Live Mode

You will now monitor real network traffic.

Tasks

- Run Suricata on your selected network interface
- Ensure it is actively capturing traffic

Deliverables

- Confirmation that Suricata is running
- Interface used

Mission 4: Generate Suspicious Traffic

Simulate attacker behavior from another machine or VM.

Tasks

Perform at least **2 of the following actions**:

- Network scan (e.g. discovering open ports)
- ICMP flood (ping flood)
- Repeated connection attempts

Deliverables

- Description of actions performed
- Source of the attack (machine used)

Mission 5: Detect and Analyze Alerts

Observe Suricata in real time.

Tasks

- Identify alerts triggered by your actions
- Analyze:
 - Type of alert
 - Protocol involved
 - Timestamp
- Compare behavior with Part 1 (PCAP analysis)

Deliverables

- Screenshot or log extract
- Explanation of detected behavior

Part 3 : Correlation & Understanding

Tasks

- Compare what you **generated** vs what Suricata **detected**
- Identify any:
 - Missed detections
 - Unexpected alerts

Deliverables

- Short analysis:
 - What was expected?
 - What was actually detected?

Tips & Best Practices

- Start with **simple attacks** before complex scenarios
 - Use **clear and isolated tests** (one action at a time)
 - If no alert appears:
 - Check rules
 - Check interface
 - Check traffic visibility
 - Logs are your **primary source of truth**
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Reflection Questions

- What is the difference between analyzing a **PCAP file** and monitoring **live traffic**?
- Why might Suricata fail to detect certain attacks?
- What is the importance of **rules tuning** in intrusion detection?
- How can false positives impact a security analyst's work?
- What additional tools could complement Suricata in a real SOC environment?

“You don’t stop attacks by blocking them; you stop attacks by understanding them”